

Sales Data Analysis using Power BI

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Tool Used: Microsoft Power BI Desktop

Project Summary

This project focuses on transforming and analyzing sales data using Microsoft Power BI. Multiple datasets were imported, cleaned, transformed, and modeled to create a structured analytical dataset. Power Query was used for data preparation, while DAX calculations supported advanced business analysis. An interactive dashboard was developed to visualize sales performance, profitability, target achievement, and regional trends. The project enables organizations to make informed business decisions through data-driven insights.

Project Objectives

- Import multiple datasets into Power BI.
 - Perform data cleaning and transformation.
 - Create calculated columns and DAX measures.
 - Build relationships between multiple tables.
 - Analyze sales performance and profitability.
 - Compare actual sales with monthly targets.
 - Design an interactive dashboard for business reporting.
 - Generate meaningful business insights for decision-making.
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Dataset Description

Dataset	Description
List of Orders	Customer information, City, State, Order Date
Order Details	Product Category, Quantity, Amount, Profit
Sales Target	Monthly category-wise sales targets

Software & Tools Used

Tool	Purpose
Power BI Desktop	Dashboard Development
Power Query	Data Cleaning & Transformation
DAX	Calculated Columns & Measures

Data Cleaning Process

1. Imported all source datasets.
 2. Retained the first 500 records for analysis.
 3. Corrected data types for dates and numeric fields.
 4. Standardized customer names using Capitalize Each Word.
 5. Created a Location column by combining City and State.
 6. Calculated Profit Margin.
 7. Created a Profit Status column.
 8. Merged datasets using Order ID.
 9. Checked for missing values.
 10. Removed duplicate records.
 11. Applied state-based filtering.
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Data Modeling

A one-to-many relationship was created between the **List of Orders** and **Order Details** tables using the **Order ID** field. The **Sales Target** table was linked to support target analysis. Proper relationships enabled accurate filtering, calculations, and dashboard interactions.

DAX Calculations

Category Type (Calculated Column)

Category Type =
'Order Details'[Category] & " - " &
'Order Details'[Sub-Category]

Revenue per Order

Revenue per Order =
Amount * Quantity

Sales Category

IF(Amount > Average Amount,
"Above Average",
"Below Average")

Measures Created

- Total Sales
 - Total Profit
 - Order Count
 - Average Profit
 - Total Target
 - YTD Sales
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Dashboard Visualizations

Sales Target Achievement by Category

A column chart compares actual sales with sales targets for each product category. This visualization helps evaluate business performance against planned objectives.

Monthly Sales Trend

A line chart displays monthly sales performance and identifies seasonal trends and peak sales periods.

Profit by Sub-Category

A clustered bar chart compares profit across different sub-categories, highlighting the highest and lowest performing products.

Sales Performance Matrix

A matrix visual summarizes sales, profit, and quantity across categories and sub-categories for detailed performance analysis.

Geographic Sales Analysis

A map visualization displays sales performance across cities and states, helping identify strong and weak sales regions.

Sales Distribution by Sub-Category

A treemap visual illustrates the contribution of each sub-category to overall sales.

Order Count Analysis by State

A funnel chart compares order volumes across different states and highlights areas with higher customer activity

Dashboard Overview

The dashboard provides a comprehensive analysis of sales performance by combining KPI cards, trend charts, category analysis, geographic visuals, and target comparisons. Interactive filters enable users to explore data by category, product, month, and region, supporting efficient business analysis.

Descriptive Analytics (What Happened?)

This project summarizes historical sales performance using interactive dashboard visualizations. Total sales, total profit, order count, monthly trends, and category-wise analysis provide a clear understanding of business performance

Diagnostic Analytics (Why Did It Happen?)

The dashboard identifies the factors influencing business performance by analyzing profitability across categories, regional sales variations, and customer purchasing

behavior. These insights explain why certain products and locations perform better than others.

Predictive Analytics (What Might Happen Next?)

Historical sales trends and target comparisons help estimate future business performance. The analysis supports demand forecasting, inventory planning, and future sales target setting.

Prescriptive Analytics (What Should Be Done?)

The project recommends focusing on high-profit products, improving performance in underperforming regions, optimizing inventory, and continuously monitoring sales targets to improve overall business performance.

Project Conclusion

This project successfully transformed raw sales data into a structured analytical model using Microsoft Power BI. Through data cleaning, transformation, modeling, DAX calculations, and dashboard development, meaningful business insights were generated. The interactive dashboard supports informed decision-making by enabling users to monitor sales performance, profitability, and regional trends effectively.

Thank You

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